The Transversal Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: PAIRS, ALGEBRA, TRIPOLY and PROVE (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (PAIRS)
- ☐ **A2** (PAIRS)
- ☐ **A3** (TRIPOLY)
- ☐ **A4** (TRIPOLY)
- ☐ **A5** (PROVE)
- ☐ **A6** (ALGEBRA)

Entrées

10 pts each · choose 4

- ☐ **E1** (ALGEBRA)
- ☐ **E2** (ALGEBRA)
- ☐ **E3** (TRIPOLY)
- ☐ **E4** (TRIPOLY)
- ☐ **E5** (PAIRS)
- ☐ **E6** (PROVE)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (PROVE)
- ☐ **S2** (ALGEBRA)
- ☐ **S3** (TRIPOLY)
- ☐ **S4** (PAIRS)

My skill check (circle each skill you have covered): PAIRS · ALGEBRA · TRIPOLY · PROVE

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 PAIRS ☐ My pick · 5 pts

Parallel lines are cut by a transversal. Two angles lie **between** the parallel lines on **opposite sides** of the transversal, and one measures 72°. Name this pair and find the other angle.

A2 PAIRS ☐ My pick · 5 pts

Parallel lines are cut by a transversal. Two same-side interior angles are shown; one measures 116°. Find the other angle and explain why.

A3 TRIPOLY ☐ My pick · 5 pts

Find the sum of the interior angles of a pentagon.

A4 TRIPOLY ☐ My pick · 5 pts

In a triangle, the two remote interior angles measure 35° and 48°. Find the exterior angle at the third vertex.

A5 PROVE ☐ My pick · 5 pts

Lines a and b are cut by a transversal, and a pair of alternate interior angles both measure 58°. Are a and b parallel? Name the theorem that tells you.

A6 ALGEBRA ☐ My pick · 5 pts

Parallel lines are cut by a transversal. Two corresponding angles measure $(3x + 10)°$ and 70°. Find x.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 ALGEBRA ☐ My pick · 10 pts

Parallel lines are cut by a transversal. Same-side interior angles measure (4x + 8)° and (2x + 10)°. Find x and both angle measures.

E2 ALGEBRA ☐ My pick · 10 pts

Parallel lines are cut by a transversal. Alternate interior angles measure (5x − 12)° and (3x + 14)°. Find x and the measure of each angle.

E3 TRIPOLY ☐ My pick · 10 pts

An exterior angle of a triangle measures (7x − 5)°. The remote interior angles measure (2x + 15)° and (3x + 10)°. Find x and the exterior angle.

E4 TRIPOLY ☐ My pick · 10 pts

Each interior angle of a regular polygon measures 156°. How many sides does the polygon have? Show your method.

E5 PAIRS ☐ My pick · 10 pts

Lines m and n are parallel and cut by a transversal. ∠2 and ∠7 are alternate exterior angles and m∠2 = 125°. Lee says m∠7 = 55° “because angles along a transversal add to 180°.” Find his mistake and give the correct m∠7.

E6 PROVE ☐ My pick · 10 pts

Lines a and b are cut by a transversal. Alternate interior angles measure (3x + 20)° and (5x − 40)°. Find x so that a and b are parallel, and justify your answer with a theorem.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 PROVE ☐ My pick · 20 pts

Lines a and b are cut by transversal t. ∠1 and ∠3 are alternate exterior angles and ∠1 ≅ ∠3. ∠2 is vertical to ∠1, and ∠2 and ∠3 are corresponding angles. Write a proof, with a reason for each step, that a and b are parallel.

S2 ALGEBRA ☐ My pick · 20 pts

Lines m and n are parallel and cut by transversal t. Alternate interior angles ∠1 and ∠2 measure (3x + 7)° and (5x − 25)°. **(a)** Find x and m∠1. **(b)** ∠3 forms a linear pair with ∠2, and ∠4 corresponds to ∠3 with m∠4 = (5y − 25)°. Find m∠3, y, and m∠4.

S3 TRIPOLY ☐ My pick · 20 pts

In triangle ABC, m∠A = (2x + 10)°, m∠B = (3x − 5)°, m∠C = (x + 25)°. **(a)** Find x and all three angles. **(b)** Extend CB past B to point D. Find the exterior angle ∠ABD in **two different ways** and show they agree.

S4 PAIRS ☐ My pick · 20 pts

Parallel lines m and n are cut by a transversal. At m the angles are ∠1 (upper left), ∠2 (upper right), ∠3 (lower left), ∠4 (lower right); at n they are ∠5, ∠6, ∠7, ∠8 in the same positions. m∠1 = 130°. **(a)** Find all eight angles. **(b)** Name the pair type for ∠1 and ∠8, ∠4 and ∠5, and ∠3 and ∠5.